

Day 16 of training

React Component

Definition:

A **React Component** is a reusable and independent piece of UI in a React application. It can display content, accept inputs (called props), maintain its own data (called state), and be combined with other components to build complex user interfaces.

Key Points:

1. Components are the **building blocks** of a React app.
2. Components make the code **modular, reusable, and easy to maintain**.
3. Components can **receive data from parent components** using **props**.
4. Components can **manage dynamic data** using **state**.
5. Components can **contain other components**, forming a **component tree**.
6. Components can respond to **user actions** like clicks, typing, etc.

Types of React Components

1. Functional Component:

- Defined as a **JavaScript function**.
- Returns **JSX** (HTML-like code) to render UI.
- Simple and easy to write.
- Can use **hooks** like `useState` for state management.

Eg

```
function Welcome(props) {  
  
  return <h1>Hello, {props.name}</h1>;  
  
}
```

2. Class Component:

- Defined as a **JavaScript class**.
- Extends `React.Component`.
- Must have a **render() method** that returns JSX.
- Can manage state without hooks.

```
class Welcome extends React.Component {  
  
  render() {  
  
    return <h1>Hello, {this.props.name}!</h1>;  
  
  }  
  
}
```

Advantages of Components:

1. **Reusability:** Write once, use many times.
2. **Maintainability:** Easy to manage code in small pieces.
3. **Separation of Concerns:** Each component does a single task.
4. **Faster Development:** Components can be tested individually.
5. **Dynamic UI:** State and props allow interactive interfaces.